HJB Derivation, Optimal Policy, and Verification for CIR–Heston with HARA and Dynamic Benchmark

Chong Han

August 31, 2025

1 Model and Objective

State variables. We consider the state

$$(t, R_t, V_t, W_t, M_t),$$

where R_t is the *short rate*, V_t the *instantaneous variance* of the risky asset, W_t the *wealth*, and M_t the *contribution benchmark*. Define the *gap* $X_t := W_t - M_t$.

Short rate (CIR).

$$dR_t \;=\; \kappa_r(\theta_r - R_t)\,dt \;+\; \sigma_r\sqrt{R_t}\,dZ_t^r. \tag{1}$$

- **Parameters:**
  - R_t: instantaneous short rate.
  - $\kappa_r > 0$: mean-reversion speed.
  - $\theta_r > 0$: long-run equilibrium interest rate.
  - $\sigma_r > 0$: volatility of rate shocks.
  - dZ_t^r: Brownian motion.
- **Economic Role:**
  - Keeps $r_t \geq 0$ (Feller condition).
  - Rates revert to θ_r, mirroring central-bank policy.
  - Volatility rises with r_t, reflecting turbulent high-rate regimes.

- **Why in our model?** Provides realistic stochastic risk-free returns, which directly affect the wealth dynamics and optimal allocation.

Variance (CIR/Heston).

$$dV_t \;=\; \kappa_v(\theta_v - V_t)\,dt \;+\; \sigma_v\sqrt{V_t}\,dZ_t^v. \tag{2}$$

Risky asset (Heston return).

$$\frac{dS_t}{S_t} \;=\; \mu_S\,dt \;+\; \sqrt{V_t}\,dW_t^S. \tag{3}$$

- **Parameters:**
  - S_t: risky asset price.
  - μ_S: constant expected return (avoid adding dimensions to 5 of HJb and make sure it's solvable and clean).
  - V_t: stochastic variance process.
  - $\kappa_v > 0$: mean-reversion speed of variance.
  - $\theta_v > 0$: long-run variance.
  - $\sigma_v > 0$: volatility of volatility.
  - dW_t^S, dZ_t^v: Brownian motions (possibly correlated).
- **Economic Role:**
  - Captures volatility clustering (empirically observed in equities).
  - Ensures variance stays nonnegative.
  - Provides realistic risk dynamics for comparative statics.
- **Why in our model?** Equity volatility is stochastic, so regret is shaped not only by wealth and benchmark but also by volatility shocks.

Correlations. Brownian motions have constant correlations

$$d\langle Z^r, Z^v\rangle_t = \rho_{rv}\,dt, \qquad d\langle W^S, Z^r\rangle_t = \rho_{Sr}\,dt, \qquad d\langle W^S, Z^v\rangle_t = \rho_{Sv}\,dt,$$

with $|\rho_{rv}|, |\rho_{Sr}|, |\rho_{Sv}| \leq 1$.

Refer to model notes page 22

Control. The control is the *dollar amount* in the risky asset, π_t (equivalently, the risky *fraction* $\alpha_t = \pi_t / W_t$).

Dynamic benchmark (contributions). We use the performance-based rule

$$dM_t \;=\; \big[c_0 + \eta\,(W_t - M_t)^+ - \xi\,(M_t - W_t)^+\big]\,dt \quad \text{with} \quad M_0 \geq 0, \tag{4}$$

where $x^+ = \max(x, 0)$.

- **Parameters:**
  - M_t: cumulative net contributions (benchmark).
  - $c_0 \geq 0$: baseline contribution rate.
  - $\eta \geq 0$: upward sensitivity if $W_t > M_t$ (optimism).
  - $\xi \geq 0$: downward sensitivity if $M_t > W_t$ (pessimism).
- **Economic Role:**
  - Contributions adapt endogenously to performance.
  - If investor is ahead ($W_t > M_t$): contributions rise.
  - If behind ($M_t > W_t$): contributions fall.
- **Why in our model?** Benchmark drives regret aversion explicitly — regret is tied to $W_t - M_t$.

Wealth dynamics. The portfolio is self-financing with external flow dM_t credited to the account:

$$dW_t \;=\; \big[R_t(W_t - \pi_t) + \mu_S \pi_t\big]\,dt \;+\; \sqrt{V_t}\,\pi_t\,dW_t^S \;+\; dM_t. \tag{5}$$

- α_t: fraction of wealth in risky asset.
- $r_t(1 - \alpha_t)W_t$: return from safe asset.
- $\mu_S \alpha_t W_t$: expected return from risky asset.
- $-dM_t$: contribution outflow to the benchmark.
- $\sqrt{v_t}\,\alpha_t W_t\,dW_t^S$: random diffusion from equity risk.

Preferences (HARA in the gap). For $a > 0$, $b \geq 0$, and $\gamma \neq 1$,

$$U(X) \;=\; \frac{[aX + b]^{1-\gamma}}{1-\gamma}, \qquad \text{domain } \{X : \; aX + b > 0\}. \tag{6}$$

- **Parameters:**
  - $W_t - M_t$: wealth relative to benchmark (gap triggers regret).
  - γ: risk-aversion parameter (HARA form).
  - a, b: scaling and shift constants.
- **Economic Role:**
  - Captures hyperbolic absolute risk aversion.
  - Makes regret explicit: utility collapses when $W_t \approx M_t$.
  - Homotheticity makes the HJB tractable.

Value function.

$$J(t, r, v, w, m) \;=\; \sup_{\pi} \; \mathbb{E}_{t,r,v,w,m}\big[U\big(W_T - M_T\big)\big]. \tag{7}$$

Terminal condition: $J(T, r, v, w, m) = U(w - m)$.

Assumption 1 (Well-posedness baseline)**.** *$\kappa_r, \theta_r, \sigma_r, \kappa_v, \theta_v, \sigma_v > 0$ satisfy Feller conditions $2\kappa_r\theta_r \geq \sigma_r^2$ and $2\kappa_v\theta_v \geq \sigma_v^2$; correlations are constant; μ_S is constant; $c_0, \eta, \xi \geq 0$. Initial state (R_0, V_0, W_0, M_0) lies in the natural domain with $a(W_0 - M_0) + b > 0$.*

Definition 1 (Admissible control)**.** *A progressively measurable π is admissible if it satisfies either a fraction constraint $\alpha_t = \pi_t / W_t \in [0, 1]$ (with $\pi_t \equiv 0$ when $W_t = 0$) or a linear-growth bound $|\pi_t| \leq K(1 + W_t)$ a.s. for some $K > 0$, and the state SDEs admit a unique strong solution with*

$$\mathbb{E}\left[\sup_{s \leq T}\big(|W_s|^p + M_s^p + R_s^p + V_s^p\big)\right] < \infty \quad \textit{for some } p > 1.$$

2 Itô Preliminaries and Quadratic Variations

Lemma 1 (Itô for $J(t,R,V,W,M)$)**.** *If* $J \in C^{1,2,2,2,2}$*, then*

$$\begin{aligned} dJ &= J_t\,dt + J_r\,dR + J_v\,dV + J_w\,dW + J_m\,dM \\ &\quad + \tfrac{1}{2}J_{rr}\,(dR)^2 + \tfrac{1}{2}J_{vv}\,(dV)^2 + \tfrac{1}{2}J_{ww}\,(dW)^2 \\ &\quad + J_{rv}\,dR\,dV + J_{rw}\,dR\,dW + J_{vw}\,dV\,dW. \end{aligned}$$

With (1)–(5), we have the quadratic covariations

$$\begin{gathered} (dR)^2 = \sigma_r^2 R\,dt, \qquad (dV)^2 = \sigma_v^2 V\,dt, \qquad (dW)^2 = V\,\pi^2\,dt, \\ dR\,dV = \rho_{rv}\sigma_r\sigma_v\sqrt{RV}\,dt, \quad dW\,dR = \rho_{Sr}\sigma_r\sqrt{RV}\,\pi\,dt, \quad dW\,dV = \rho_{Sv}\sigma_v V\,\pi\,dt. \end{gathered} \tag{8}$$

Substitute dR, dV, dW, dM into dJ and collect dt terms. Write the drifts and diffusions explicitly:

$$\begin{gathered} dR = \underbrace{\kappa_r(\theta_r - R)}_{\mu_R}\,dt + \underbrace{\sigma_r\sqrt{R}}_{\sigma_R}\,dZ^r, \quad dV = \underbrace{\kappa_v(\theta_v - V)}_{\mu_V}\,dt + \underbrace{\sigma_v\sqrt{V}}_{\sigma_V}\,dZ^v, \\ dW = \underbrace{R(W-\pi) + \mu_S\pi}_{\mu_W}\,dt + \underbrace{\sqrt{V}\,\pi}_{\sigma_W}\,dW^S + \underbrace{\dot{M}}_{\mu_M}\,dt, \quad dM = \dot{M}\,dt. \end{gathered}$$

Thus the *drift* (the dt-coefficient) of dJ is

$$\begin{aligned} \mathrm{Drift}(dJ) \;=\; & J_t + \mu_R J_r + \tfrac{1}{2}\sigma_R^2 J_{rr} + \mu_V J_v + \tfrac{1}{2}\sigma_V^2 J_{vv} + \rho_{rv}\sigma_R\sigma_V J_{rv} \\ & + \mu_W J_w + \tfrac{1}{2}\sigma_W^2 J_{ww} + \rho_{Sr}\sigma_R\sigma_W J_{rw} + \rho_{Sv}\sigma_V\sigma_W J_{vw} \\ & + \quad \underbrace{\dot{M} J_w + \dot{M} J_m}_{\text{from } dW \text{ (inflow) and } dM} \quad . \end{aligned}$$

Plug back $\mu_R, \sigma_R, \mu_V, \sigma_V, \mu_W, \sigma_W$ to obtain:

$$\begin{aligned} \mathrm{Drift}(dJ) \;=\; & J_t + \kappa_r(\theta_r - r)J_r + \tfrac{1}{2}\sigma_r^2 r\,J_{rr} + \kappa_v(\theta_v - v)J_v + \tfrac{1}{2}\sigma_v^2 v\,J_{vv} + \rho_{rv}\sigma_r\sigma_v\sqrt{rv}\,J_{rv} \\ & + \big[r(w-\pi) + \mu_S\pi\big]J_w + \tfrac{1}{2}v\,\pi^2\,J_{ww} + \rho_{Sr}\sigma_r\sqrt{rv}\,\pi\,J_{rw} + \rho_{Sv}\sigma_v v\,\pi\,J_{vw} \\ & + \dot{M}\,J_w + \dot{M}\,J_m. \end{aligned}$$

3 HJB Equation and First-Order Condition

Proposition 1 (Hamilton–Jacobi–Bellman). *Under Assumptions 1 and admissible π, By the Bellman principle, the drift of J under the optimal control must vanish:*

$$0 = \sup_{\pi} \ \mathrm{Drift}(dJ),$$

Which gives the HJB:

$$\begin{aligned} 0 = \sup_{\pi} \Big\{ & J_t + \kappa_r(\theta_r - r)J_r + \tfrac{1}{2}\sigma_r^2 r\, J_{rr} + \kappa_v(\theta_v - v)J_v + \tfrac{1}{2}\sigma_v^2 v\, J_{vv} + \rho_{rv}\sigma_r\sigma_v\sqrt{rv}\, J_{rv} \\ & + \big[r(w-\pi) + \mu_S \pi\big] J_w + \tfrac{1}{2} v\, \pi^2\, J_{ww} + \rho_{Sr}\sigma_r\sqrt{rv}\, \pi\, J_{rw} + \rho_{Sv}\sigma_v v\, \pi\, J_{vw} \\ & + \dot{M}\, J_w + \dot{M}\, J_m \Big\}, \qquad J(T,r,v,w,m) = U(w-m). \end{aligned} \tag{9}$$

, with $\dot{M} := c_0 + \eta(w-m)^+ - \xi(m-w)^+$ and terminal condition $J(T,r,v,w,m) = U(w-m)$.

Remark 1 (Where the benchmark cancels in the driver). *If preferences depend on the* gap *$X = w - m$ only, write $J(t,r,v,w,m) = \tilde{J}(t,r,v,X)$. Then $J_w = \tilde{J}_X$ and $J_m = -\tilde{J}_X$, hence $\dot{M}(J_w + J_m) \equiv 0$. The benchmark still affects the* state *(hence future risk tolerance), but not the instantaneous HJB driver.*

Concavity in π. Since $v > 0$ and $J_{ww} < 0$ under risk aversion, the Hamiltonian is strictly concave in π and the supremum is attained by the FOC.

Proposition 2 (FOC and optimal dollar policy). *Group the π-dependent terms in the Hamiltonian:*

$$\mathcal{H}(\pi) = \underbrace{(\mu_S - r)J_w}_{\textit{Sharpe channel}}\,\pi + \tfrac{1}{2} v\, J_{ww}\, \pi^2 + \underbrace{\rho_{Sr}\sigma_r\sqrt{rv}\, J_{rw}}_{\textit{term-structure hedge}}\,\pi + \underbrace{\rho_{Sv}\sigma_v v\, J_{vw}}_{\textit{volatility hedge}}\,\pi + (\textit{terms independent of } \pi).$$

FOC: $\partial\mathcal{H}/\partial\pi = 0$ gives

$$(\mu_S - r)J_w + v\,\pi\, J_{ww} + \rho_{Sr}\sigma_r\sqrt{rv}\, J_{rw} + \rho_{Sv}\sigma_v v\, J_{vw} = 0,$$

so

$$\boxed{\pi^* \;=\; -\,\frac{(\mu_S - r)J_w + \rho_{Sr}\sigma_r\sqrt{rv}\, J_{rw} + \rho_{Sv}\sigma_v v\, J_{vw}}{v\, J_{ww}}\;.} \tag{10}$$

If portfolio constraints apply (e.g. $\alpha = \pi/W \in [0,1]$), the optimal control is the Euclidean projection of (10) onto the feasible set.

4 Homothetic Reduction (HARA-in-the-Gap)

We posit the homothetic form

$$J(t,r,v,w,m) \;=\; \frac{[a(w-m)+b]^{1-\gamma}}{1-\gamma}\, G(t,r,v), \qquad G(T,r,v)=1, \quad G>0. \tag{11}$$

Let $X := w-m$. Then

$$\begin{aligned} J_w &= a[aX+b]^{-\gamma}G, & J_{ww} &= -\gamma a^2[aX+b]^{-\gamma-1}G, \\ J_{rw} &= a[aX+b]^{-\gamma}G_r, & J_{vw} &= a[aX+b]^{-\gamma}G_v, \qquad J_m = -J_w. \end{aligned} \tag{12}$$

Definition 2 (Risk tolerance in the gap). *The (dollar) risk tolerance associated with the gap X is*

$$\mathcal{R}(X) \;:=\; -\frac{J_w}{J_{ww}} \;=\; \frac{aX+b}{\gamma a}.$$

Proposition 3 (Closed-form feedback control). *Substituting* (12) *into* (10) *and simplifying gives*

$$\boxed{\pi^*(t,r,v,X) \;=\; \mathcal{R}(X)\left[\frac{\mu_S-r}{v} + \rho_{Sr}\sigma_r\sqrt{\frac{r}{v}}\,\frac{G_r}{G} + \rho_{Sv}\sigma_v\,\frac{G_v}{G}\right].} \tag{13}$$

The risky fraction is $\alpha^* = \pi^*/W$ (project to $[0,1]$ if constrained).

Remark 2 (Benchmark proximity amplifies risk aversion). *As $X \downarrow 0$ (wealth approaches benchmark), $\mathcal{R}(X)\downarrow$ and the dollar risky position shrinks mechanically. This is the direct, instantaneous regret channel. Intertemporally, M_t also shifts the drift of X_t, altering future $\mathcal{R}(X)$ and thus future allocations.*

5 Reduced PDE for the Multiplier G

Proposition 4 (Value equation for G). *Under the gap ansatz* (11) *and the benchmark cancellation in the driver, the HJB* (9) *reduces to the semilinear PDE. In other words: Insert the ansatz and π^* back into the HJB. Factor out $[aX+b]^{1-\gamma}$ and use the cancellation $\dot M(J_w+J_m)\equiv 0$. After simplification, G solves:*

$$\begin{aligned} 0 \;=\; & G_t + \kappa_r(\theta_r - r)\,G_r + \tfrac{1}{2}\sigma_r^2 r\,G_{rr} + \kappa_v(\theta_v - v)\,G_v + \tfrac{1}{2}\sigma_v^2 v\,G_{vv} + \rho_{rv}\sigma_r\sigma_v\sqrt{rv}\,G_{rv} \\ & + (1-\gamma)\,r\,G + \frac{1-\gamma}{2\gamma}\left[\frac{\mu_S-r}{\sqrt{v}} + \rho_{Sr}\sigma_r\sqrt{r}\,\frac{G_r}{G} + \rho_{Sv}\sigma_v\sqrt{v}\,\frac{G_v}{G}\right]^2 G, \end{aligned} \tag{14}$$

with terminal condition $G(T,r,v)=1$.

Remark 3 (On alternative factorizations)**.** *If one prefers to keep an explicit benchmark-to-gap ratio in the safe channel, one can factor J further to carry a bounded scalar $\zeta = \frac{a(W+M)}{aX+b}$; the qualitative results and the policy* (13) *are unchanged, while G solves a PDE in (t, r, v, ζ).*

Special Discription of How Benchmark $\Rightarrow$ Policy

1: Put the benchmark where preferences live (the gap). We model regret through the *gap* $X_t := W_t - M_t$ and adopt HARA-in-the-gap utility

$$U(X) \;=\; \frac{[aX+b]^{\,1-\gamma}}{1-\gamma}, \qquad a>0,\; b\geq 0,\; \gamma>0,\; \gamma\neq 1,$$

so the value function factors homothetically as

$$J(t,r,v,w,m) \;=\; \frac{[a(w-m)+b]^{\,1-\gamma}}{1-\gamma}\, G(t,r,v).$$

Why this matters: the benchmark M enters through $X = w - m$ and *rescales* both marginal utility and risk tolerance; G captures *intertemporal* effects driven by (R_t, V_t).

2: Direct (static) link to the optimal π^* . From the HJB and first-order condition, the optimal *dollar* risky position is

$$\boxed{\pi^*(t,r,v,X) \;=\; \underbrace{\mathcal{R}(X)}_{\text{risk tolerance in gap}} \left[\frac{\mu_S - r}{v} + \rho_{Sr}\sigma_r\sqrt{\tfrac{r}{v}}\,\frac{G_r}{G} + \rho_{Sv}\sigma_v\,\frac{G_v}{G}\right],} \tag{15}$$

where the *gap risk tolerance* is

$$\mathcal{R}(X) \;:=\; -\frac{J_w}{J_{ww}} = \frac{aX+b}{\gamma a} \;=\; \frac{X}{\gamma} + \frac{b}{\gamma a}.$$

Immediate implication (benchmark proximity): as $X \downarrow 0$ (wealth approaches the benchmark), $\mathcal{R}(X)$ drops toward $\frac{b}{\gamma a}$ (or to 0 if $b = 0$). Thus the *entire* risky dollar demand in (15) is proportionally *suppressed.*

3: Local sensitivity in the gap (clear comparative statics-TO DO NEXT). Holding (t,r,v) fixed, G does not depend on X. Hence

$$\frac{\partial\pi^*}{\partial X} \;=\; \underbrace{\frac{\partial\mathcal{R}}{\partial X}}_{=1/\gamma}\left[\frac{\mu_S - r}{v} + \rho_{Sr}\sigma_r\sqrt{\tfrac{r}{v}}\,\frac{G_r}{G} + \rho_{Sv}\sigma_v\,\frac{G_v}{G}\right].$$

Interpretation: per *dollar* that wealth moves away from the benchmark (higher X), optimal risky *dollars* increase by exactly the bracket divided by risk aversion γ. With zero correlations

($\rho_{Sr} = \rho_{Sv} = 0$) this collapses to

$$\frac{\partial \pi^*}{\partial X} \;=\; \frac{1}{\gamma}\,\frac{\mu_S - r}{v},$$

a transparent rule: benchmark proximity linearly tilts the risky position.

4: From dollars to fraction. The risky *fraction* $\alpha^* = \pi^*/W$ inherits the gap dependence:

$$\alpha^*(t, r, v, W, M) \;=\; \frac{1}{W}\Big(\tfrac{X}{\gamma} + \tfrac{b}{\gamma a}\Big)\left[\frac{\mu_S - r}{v} + \rho_{Sr}\sigma_r\sqrt{\tfrac{r}{v}}\,\frac{G_r}{G} + \rho_{Sv}\sigma_v\,\frac{G_v}{G}\right].$$

So, for a fixed gap X, larger W dilutes the fraction (classic scale effect), while for a fixed W moving X toward 0 reduces equity exposure (the regret effect).

5: Dynamic (intertemporal) link via the X–SDE. Wealth includes the flow dM_t, so $dX_t = dW_t - dM_t$ cancels the *instantaneous* contribution flow:

$$dX_t \;=\; \big[r_t(W_t - \pi_t) + \mu_S \pi_t\big]\,dt + \sqrt{V_t}\,\pi_t\,dW_t^S.$$

Equivalently, using $W_t = X_t + M_t$,

$$dX_t \;=\; \big[r_t X_t + r_t(M_t - \pi_t) + \mu_S \pi_t\big]\,dt + \sqrt{V_t}\,\pi_t\,dW_t^S.$$

Implications:

- The benchmark rule $dM_t = \big[c_0 + \eta(W_t - M_t)^+ - \xi(M_t - W_t)^+\big]dt$ does not appear explicitly in the diffusion or instantaneous drift of X_t (it cancels out at the dt level), but M_t still enters the *drift* through $r_t M_t$ because $W_t = X_t + M_t$.

- The gap process thus inherits rate and equity risks through π_t, and the *level* of M_t tilts the drift. Since π^* scales with $\mathcal{R}(X_t)$, the benchmark indirectly governs future risk taking by moving X_t and hence future risk tolerance.

Step 6: Where the benchmark enters the value (and hedging) channel. While the HJB driver has the cancellation $\dot{M}(J_w + J_m) \equiv 0$ (because $J_w = J_X$ and $J_m = -J_X$), the benchmark still affects:

(a) the level effect: the terminal condition $J(T, \cdot) = U(W_T - M_T)$, hence the initial value $J(t, \cdot)$ via X;

(b) the policy effect: π^* scales with $\mathcal{R}(X)$, shrinking (expanding) risky dollars near (far from) the benchmark;

(c) the hedging effect: the intertemporal hedging terms $\frac{G_r}{G}, \frac{G_v}{G}$ are multiplied by $\mathcal{R}(X)$, so benchmark proximity proportionally damps hedging demand as well.

Solving the G–PDE quantifies these channels and yields the comparative statics we want: as $X \downarrow 0$, both the *myopic* and the *hedging* components of (15) are compressed.

Step 7: Minimal "toy" rule for intuition. If $\rho_{Sr} = \rho_{Sv} = 0$ and we freeze (r, v) over a short horizon,

$$\boxed{\pi^* \approx \left(\tfrac{X}{\gamma} + \tfrac{b}{\gamma a}\right)\frac{\mu_S - r}{v}, \qquad \alpha^* \approx \frac{1}{W}\left(\tfrac{X}{\gamma} + \tfrac{b}{\gamma a}\right)\frac{\mu_S - r}{v}.}$$

Read: every \$1 that wealth sits above the benchmark adds approximately $\frac{1}{\gamma}\frac{\mu_S - r}{v}$ dollars of risky asset; every \$1 closer to the benchmark removes the same amount.

Key takeaways (what to remember).

- The benchmark matters *twice*: (i) directly, via the gap risk tolerance $\mathcal{R}(X)$ that scales *all* risky demand, and (ii) dynamically, by steering the drift of X_t (through M_t inside W_t) and therefore tomorrow's risk tolerance.
- Near the benchmark ($X \approx 0$), both myopic demand and intertemporal hedging demand are *systematically lower.* This is the precise, model-based manifestation of how benchmark proximity amplifies risk aversion.

6 Verification and Well-posedness

Assumption 2 (Regularity and integrability). *$G \in C^{1,2,2}\big([0,T]\times(0,\infty)\times(0,\infty)\big)$ is positive, of at most polynomial growth; the feedback π^* in* (13) *(or its projection onto constraints) is admissible; and J in* (11) *is $C^{1,2,2,2,2}$ with polynomial growth on its domain $\{a(w-m)+b > 0\}$.*

Theorem 1 (Verification). *Under Assumptions 1 and 2:*

(i) For any admissible π, the process $Y_s := J(s, R_s, V_s, W_s, M_s)$ is a supermartingale. Hence $J \geq V$, where V is the value function in (7).

(ii) For the feedback control π^* in (13) *(or its projected version under constraints), Y_s is a martingale and $J = V$. In particular, π^* is optimal.*

Sketch. Apply Itô (Lemma 1) to J along any admissible state/control and use (8). The drift equals $J_t + \mathcal{H}(x, \pi, \nabla J, \nabla^2 J)$, where $\mathcal{H}$ is the Hamiltonian. Since J solves the HJB with supremum over π, $J_t + \sup_{\tilde{\pi}} \mathcal{H} = 0$, the drift is $-\big(\sup_{\tilde{\pi}} \mathcal{H} - \mathcal{H}(x, \pi, \cdot)\big) \leq 0$, giving supermartingale property and $J \geq V$. For π^* that attains the supremum (strict concavity in π), the drift is zero and Y is a martingale; taking expectations with $J(T, \cdot) = U(\cdot)$ yields $J = V$. Integrability follows from polynomial growth and admissibility (localization if needed). □

Remark 4 (Well-posedness of the state SDEs). *Under Feller conditions the CIR boundaries $r = 0$ and $v = 0$ are non-attainable (or instantaneously reflecting in the limit), and moments of (R, V) are finite on $[0, T]$. With admissible π, wealth has linear-growth coefficients and finite moments. Thus, solutions exist uniquely and do not explode on finite horizons.*

7 Economic Decomposition of the Optimal Policy

The feedback (13) splits into:

$$\underbrace{\mathcal{R}(X)\,\frac{\mu_S - r}{v}}_{\text{Merton/myopic demand}} \quad + \quad \underbrace{\mathcal{R}(X)\left[\rho_{Sr}\sigma_r\sqrt{\tfrac{r}{v}}\,\tfrac{G_r}{G} + \rho_{Sv}\sigma_v\,\tfrac{G_v}{G}\right]}_{\text{intertemporal hedging against } R,V}.$$

- *Regret channel:* $\mathcal{R}(X) = (aX + b)/(\gamma a)$ shrinks as $X = W - M$ approaches 0, suppressing the *dollar* exposure near the benchmark.
- *Hedging channel:* G_r/G and G_v/G are factor-elasticities of the continuation value; correlations ρ_{Sr}, ρ_{Sv} scale their importance. If $\rho_{Sr} = \rho_{Sv} = 0$, hedging terms vanish.

8 Implementation Notes

- **Constraints:** If $\alpha \in [0,1]$, replace (13) by its projection $\Pi_{[0,W]}(\pi^*)$; the verification holds with a variational inequality HJB.
- **Numerics:** Solve (14) on (t,r,v) with terminal $G(T,\cdot)=1$ (FD/FE, sparse grids). Comparative statics in $(X,\gamma,\eta,\xi,\rho_{\cdot})$ follow directly from (13).
- **Long-run limits:** If $R_t \to \theta_r$ and $V_t \to \theta_v$ in mean, G_r/G and G_v/G stabilize and (13) tends to a stationary linear rule in X.